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REVIEW

Artificial intelligence in Finance: a comprehensive review through bibliometric and content analysis

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Abstract

Over the past two decades, artificial intelligence (AI) has experienced rapid development and is being used in a wide range of sectors and activities, including finance. In the meantime, a growing and heterogeneous strand of literature has explored the use of AI in finance. The aim of this study is to provide a comprehensive overview of the existing research on this topic and to identify which research directions need further investigation. Accordingly, using the tools of bibliometric analysis and content analysis, we examined a large number of articles published between 1992 and March 2021. We find that the literature on this topic has expanded considerably since the beginning of the XXI century, covering a variety of countries and different AI applications in finance, amongst which Predictive/forecasting systems, Classification/detection/early warning systems and Big data Analytics/Data mining /Text mining stand out. Furthermore, we show that the selected articles fall into ten main research streams, in which AI is applied to the stock market, trading models, volatility forecasting, portfolio management, performance, risk and default evaluation, cryptocurrencies, derivatives, credit risk in banks, investor sentiment analysis and foreign exchange management, respectively. Future research should seek to address the partially unanswered research questions and improve our understanding of the impact of recent disruptive technological developments on finance.

Keywords Artificial intelligence · Finance · Machine learning · Bibliometric analysis · Content analysis

JEL Classification O33 · G3

Introduction

The first two decades of the twenty-first century have experienced an unprecedented way of technological progress, which has been driven by advances in the development of cutting-edge digital technologies and applications in Artificial Intelligence

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(AI). Artificial intelligence is a field of computer science that creates intelligent machines capable of performing cognitive tasks, such as reasoning, learning, taking action and speech recognition, which have been traditionally regarded as human tasks (Frankenfield 2021). AI comprises a broad and rapidly growing number of technologies and fields, and is often regarded as a general-purpose technology, namely a technology that becomes pervasive, improves over time and generates complementary innovation (Bresnahan and Trajtenberg 1995). As a result, it is not surprising that there is no consensus on the way AI is defined (Van Roy et al. 2020). An exhaustive definition has been recently proposed by Acemoglu and Restrepo (2020, p.1), who assert that Artificial Intelligence is "(…) the study and development of intelligent (machine) agents, which are machines, software or algorithms that act intelligently by recognising and responding to their environment." Even though it is often difficult to draw precise boundaries, this promising and rapidly evolving field mainly comprises machine learning, deep learning, NLP (natural language processing) platforms, predictive APIs (application programming interface), image recognition and speech recognition (Martinelli et al. 2021).

The term "Artificial intelligence" was first coined by John McCarthy in 1956 during a conference at Dartmouth College to describe "thinking machines" (Buchanan 2019). However, until 2000, the lack of storage capability and low computing power prevented any progress in the field. Accordingly, governments and investors lost their interest and AI fell short of financial support and funding in 1974–1980 and again in 1987–1993. These periods of funding shortage are also known as "AI winters[1]".

However, the most significant development and spread of AI-related technologies is much more recent, and has been prompted by the availability of large unstructured databases, the explosion of computing power, and the rise in venture capital intended to support innovative, technological projects (Ernst et al. 2018). One of the most distinctive The term AI winter first appeared in 1characteristics of AI technologies is that, unlike industrial robots, which need to receive specific instructions, generally provided by a software, before they perform any action, can learn for themselves how to map information about the environment, such as visual and tactile data from a robot's sensors, into instructions sent to the robot's actuators (Raj and Seamans 2019). Additionally, as remarked by Ernst et al. (2018), whilst industrial robots mostly perform manual tasks, AI technologies are able to carry out activities that, until some years ago, were still regarded as typically human, i.e. what Ernst and co-authors label as "mental tasks".

The adoption of AI is likely to have remarkable implications for the subjects adopting them and, more in general, for the economy and the society. In particular, it is expected to contribute to the growth of the global GDP, which, according to a study conducted by Pricewater-house-Coopers (PwC) and published in 2017, is likely to increase by up to 14% by 2030. Moreover, companies adopting AI

[1] The term AI winter first appeared in 1984 as the topic of a public debate at the annual meeting of the American Association of Artificial Intelligence (AAAI). It referred to hype generated by over promises from developers, unrealistically high expectations from end users, and extensive media promotion.

technologies sometimes report better performance (Van Roy et al. 2020). Concerning the geographic dimension of this field, North America and China are the leading investors and are expected to benefit the most from AI-driven economic returns. Europe and emerging markets in Asia and South America will follow, with moderate profits owing to fewer and later investments (PwC 2017). AI is going to affect labour markets as well. The demand for high-skilled employees is expected to increase, whilst the demand for low-skilled jobs is likely to shrink because of automation; the resulting higher unemployment rate, however, is going to be offset by the new job opportunities offered by AI (Ernst et al. 2018; Acemoglu and Restrepo 2020).

AI solutions have been introduced in every major sector of the economy; a sector that is witnessing a profound transformation led by the ongoing technological revolution is the financial one. Financial institutions, which rely heavily on Big Data and process automation, are indeed in a "unique position to lead the adoption of AI" (PwC 2020), which generates several benefits: for instance, it encourages automation of manufacturing processes which in turn enhances efficiency and productivity. Next, since machines are immune to human errors and psychological factors, it ensures accurate and unbiased predictive analytics and trading strategies. AI also fosters business model innovation and radically changes customer relationships by promoting customised digital finance, which, together with the automation of processes, results in better service efficiency and cost-saving (Cucculelli and Recanatini 2022). Furthermore, AI is likely to have substantial implications for financial conduct and prudential supervisors, and it also has the potential to help supervisors identify potential violations and help regulators better anticipate the impact of changes in regulation (Wall 2018). Additionally, complex AI/machine learning algorithms allow Fintech lenders to make fast (almost instantaneous) credit decisions, with benefits for both the lenders and the consumers (Jagtiani and John 2018). Intelligent devices in Finance are used in a number of areas and activities, including fraud detection, algorithmic trading and high-frequency trading, portfolio management, credit decisions based on credit scoring or credit approval models, bankruptcy prediction, risk management, behavioural analyses through sentiment analysis and regulatory compliance.

In recent years, the adoption of AI technologies in a broad range of financial applications has received increasing attention by scholars; however, the extant literature, which is reviewed in the next section, is quite broad and heterogeneous in terms of research questions, country and industry under scrutiny, level of analysis and method, making it difficult to draw robust conclusions and to understand which research areas require further investigation. In the light of these considerations, we conduct an extensive review of the research on the use of AI in Finance thorough which we aim to provide a comprehensive account of the current state of the art and, importantly, to identify a number of research questions that are still (partly) unanswered. This survey may serve as a useful roadmap for researchers who are not experts of this topic and could find it challenging to navigate the extensive and composite research on this subject. In particular, it may represent a useful starting point for future empirical contributions, as it provides an account of the state of the art and of the issues that deserve further investigation. In doing so, this study complements some previous systematic reviews on the topic, such as the ones recently

conducted by Hentzen et al. (2022b) and (Biju et al. 2020), which differ from our work in the following main respects: Hentzen and co-authors' study focuses on customer-facing financial services, whilst the valuable contribution of Biju et al. poses particular attention to relevant technical aspects and the assessment of the effectiveness and the predictive capability of machine learning, AI and deep learning mechanisms within the financial sphere; in doing so, it covers an important issue which, however, is out of the scope of our work.

From our review, it emerges that, from the beginning of the XXI century, the literature on this topic has significantly expanded, and has covered a broad variety of countries, as well as several AI applications in finance, amongst which Predictive/forecasting systems, Classification /detection/early warning systems and Big data Analytics/Data mining /Text mining stand out. Additionally, we show that the selected articles can be grouped into ten main research streams, in which AI is applied to the stock market, trading models, volatility forecasting, portfolio management, performance, risk & default evaluation, cryptocurrencies, derivatives, credit risks in banks, investor sentiment analysis and foreign exchange management, respectively.

The balance of this paper is organised as follows: Sect. "Methodology" shortly presents the methodology. Sect. "A detailed account of the literature on AI in Finance" illustrates the main results of the bibliometric analysis and the content analysis. Sect. "Issues that deserve further investigation" draws upon the research streams described in the previous section to pinpoint several potential research avenues. Sect. "Conclusions" concludes. Finally, Appendix 1 clarifies some AI-related terms and definitions that appear several times throughout the paper, whilst Appendix 2 provides more information on some of the articles under scrutiny.

Methodology

To conduct a sound review of the literature on the selected topic, we resort to two well-known and extensively used approaches, namely bibliometric analysis and content analysis. Bibliometric analysis is a popular and rigorous method for exploring and analysing large volumes of scientific data which allows us to unpack the evolutionary nuances of a specific field whilst shedding light on the emerging areas in that field (Donthu et al. 2021). In this study, we perform bibliometric analysis using HistCite, a popular software package developed to support researchers in elaborating and visualising the results of literature searches in the Web of Science platform. Specifically, we employ HistCite to recover the annual number of publications, the number of forward citations (which we use to identify the most influential journals and articles) and the network of co-citations, namely, all the citations received and given by journals belonging to a certain field, which help us identify the major research streams described in Sect. "Identification of the major research streams". After that, to delve into the contents of the most pertinent studies on AI in finance, we resort to traditional content analysis, a research method that provides a systematic and objective means to make valid inferences from verbal, visual, or written data which, in turn, permit to describe and quantify specific phenomena (Downe-Wambolt 1992).